



WJEC

Chemistry A2

Unit 4

IGC HK Exam – WJEC
Condense Notes

Written by experienced author, Ian Lam, this book's engaging visual style and comprehensive detail will support you through the A2 course and help you prepare for your exams.

- This book offers high quality support
- Each topic includes condense knowledge and mark scheme answers. all written in clear uncomplicated language
- New requirements for practical work are thoroughly supported throughout

About the author:

Mr Ian Lam has nearly two years' experience for A Level Chemistry, and was trained across the world. Before writing this, Ian has done research on various topics across different subjects.

WJEC

Chemistry A2

Unit 4

Ian Lam

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Cover Image: 2023/3/17 Ian Lam

These notes have been authored by Ian Lam and are provided as support to students revising for their GCE A level exams. Though the resources are comprehensive, they may not cover every aspect of the specification and do not represent the depth of knowledge required for each unit of work.

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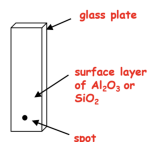
Paper Chromatography

Stationary Phase (Paper) – A solid or a liquid supported on a solid (slowing down)

Mobile Phase (Solvent) – A liquid or a gas. The mobile phase flows through the stationary phase and carries the components of mixture with it (allowing particle move)

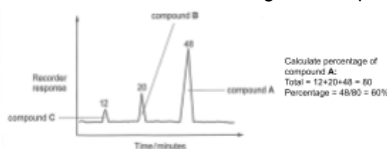
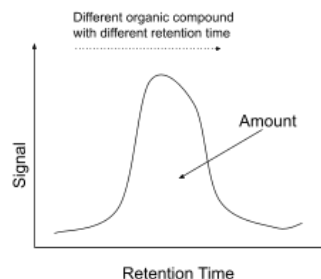
Thin Layer Chromatography (TLC)

- Separating small quantities of organic compound
- Stationary phase – silica gel or alumina on glass support
- Moving phase separates the components as it rise up



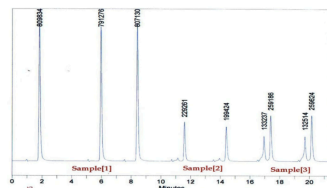
Gas Liquid Chromatography (GLC)

- Stationary Phase: gas Mobile Phase: gel
- Mobile phase and stationary phase is a high boiling point liquid
- Sample is inserted by the flow of an inert, gaseous mobile phase
- Retention time – time taken for the compound to come out of the column
- Relative area of each peak gives the percentage of each compound present in the mixture
- Different retention time shows different organic compound

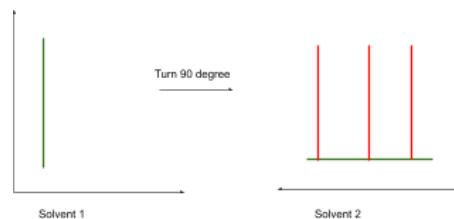


High performance Liquid Chromatography (HPLC)

- Column is packed with uniform solid particle



Two Way Chromatography



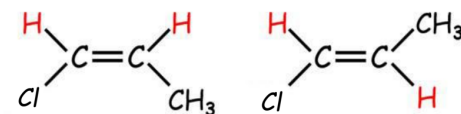
4.1 Isomerism

Stereoisomer

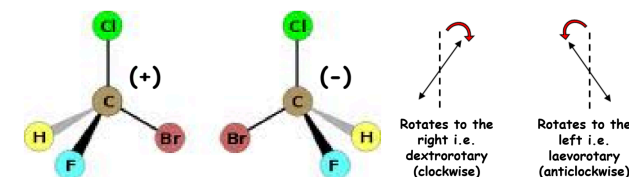
Geometric Isomer (E-Z Isomer)

z - isomer which has two higher priority substituents on the same side of the carbon-carbon double bond or planar ring plane, z.

ε - isomer which has two substituents on the opposite side of the carbon-carbon double bond or ring planar, ε.



Optical Isomer (Enantiomers)

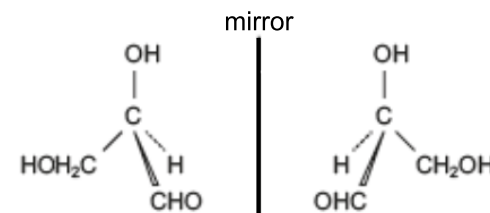


- Non-superposable mirror images → Have line of symmetry
- Molecules possesses chiral center
- Racemic Mixture → Equimolar solution of both isomers
- Different isomer will rotate the plane of polarised light to opposite direction

Chiral centre: An atom which has four different group attached

Racemic mixture: Equimolar mixture of both enantiomers. Optically inactive

Example:

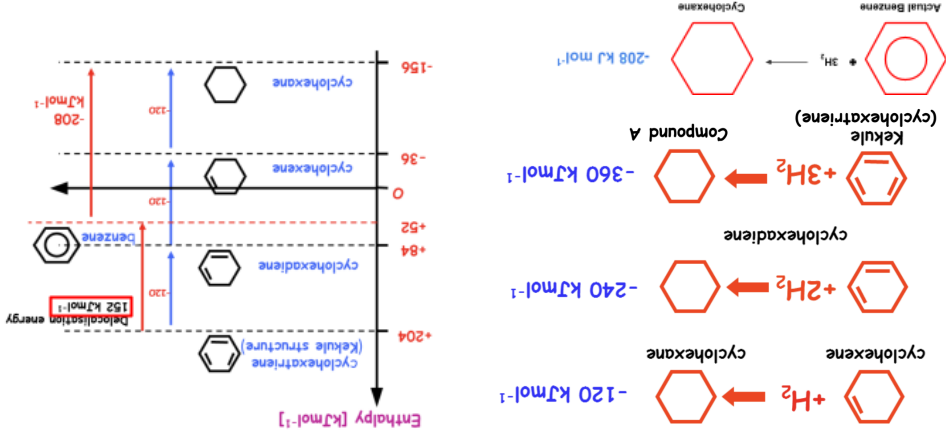


4.2 Aromaticity – Benzene



Proof of Kekulé Structure vs Actual Structure

1. Enthalpy of Hydrogenation Data



- 152 kJ mol⁻¹ is Delocalisation energy / Stabilisation energy
- **Enthalpy of hydrogenation of benzene is less exothermic than Kekulé Structure**
- There are delocalised pi (π) bonding system in Actual structure → delocation energy

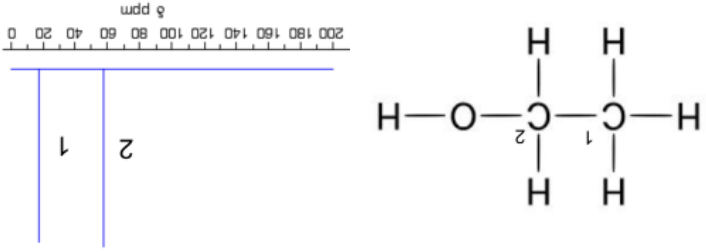
2. Mean Bond Enthalpy Data

- Actual bond enthalpy of Benzene is less than Kekulé Structure
- This suggest other types of bond are formed
- 3. Do not carry out addition reaction → Substitution only
- 4. IR Spectrum no C—C / C=C bond
- 5. Bond Length (All C—C bond length are identical in Benzene)

C—C in cyclohexane > C—C in benzene > C—C in cyclohexene

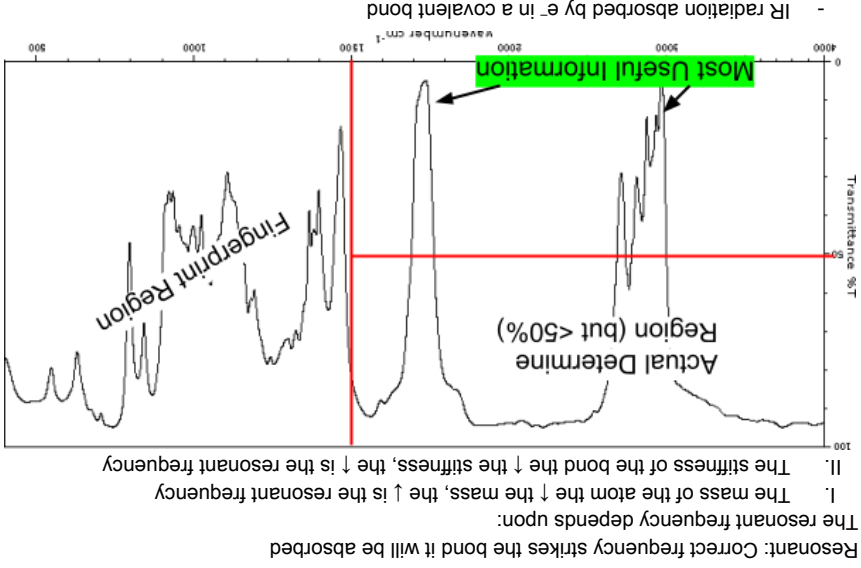
C—C cyclohexane = 0.15nm
C=C benzene = 0.14nm
C=C cyclohexene = 0.13nm

C-13 NMR Spectra (AS)

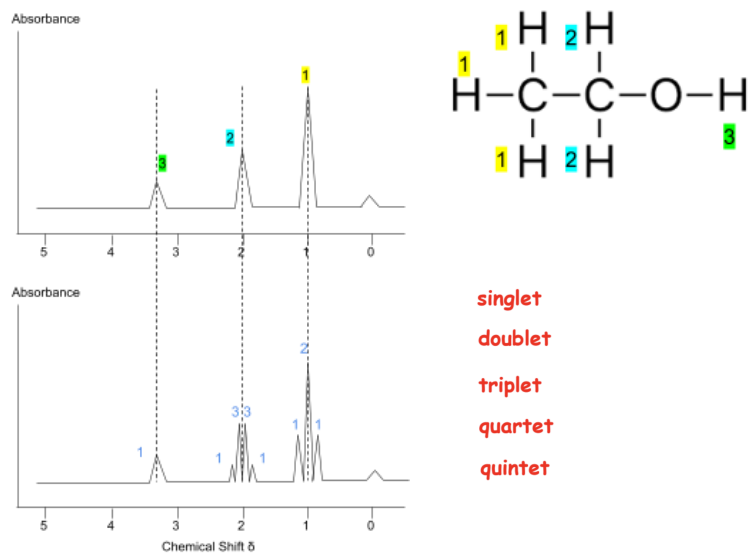


- There are 2 carbon environment so 2 peaks
- Compare the chemical shift to WJEC formula booklet (page 2)

Infrared Spectroscopy IR (AS)

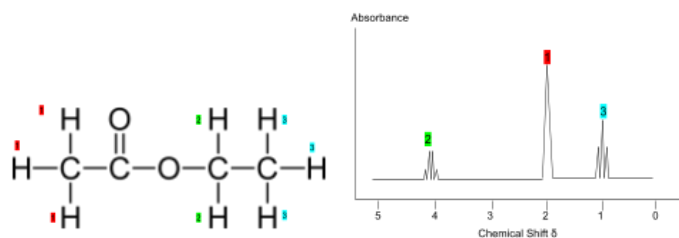


Ethanol – High Resolution NMR



- Coupling effect: Caused by the influence of hydrogen atoms bonded to neighbouring atoms
 - Proton attached to the electronegative atom (3) absorption peak will not split
 - Total area is still in same ratio
 - n + 1 rule: Each absorption will be split into n+1 peaks, where 'n' is the total number of hydrogen atoms attached to adjacent carbon atom(s).
- In other words: The number of peak is the number of Hydrogen in the adjacent carbon plus 1

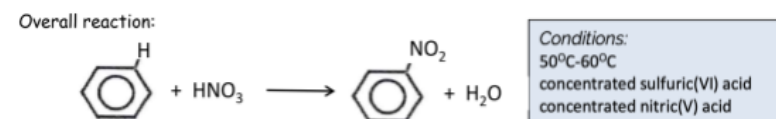
Ethyl Ethanoate – High Resolution NMR



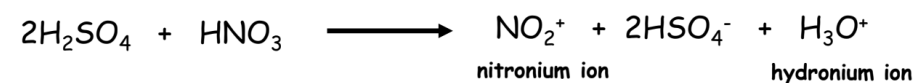
- Number 1 has no carbon on the adjacent carbon so no splitting
- Number 2 has 3 H on the adjacent carbon (which is number 3) so it has 3 + 1 peak
- Number 3 has 2 H on the adjacent carbon (which is number 2) so it has 2 + 1 peak

Reaction of Benzene — Electrophilic Substitution

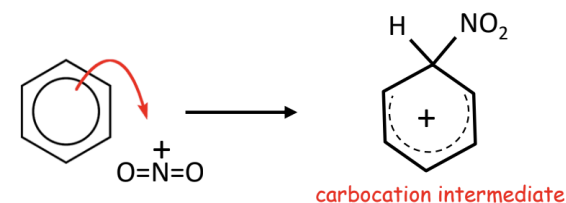
Nitration of Benzene



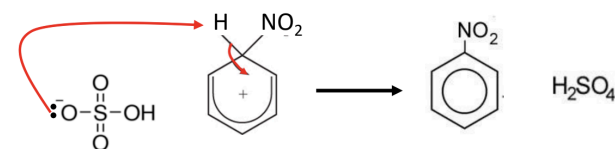
Step 1 – Generation of Electrophile



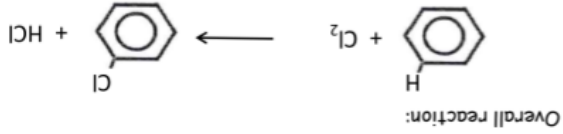
Step 2 – Attack of Electrophile on Benzene Ring



Step 3 – Abstraction of Proton & Regeneration of Catalyst



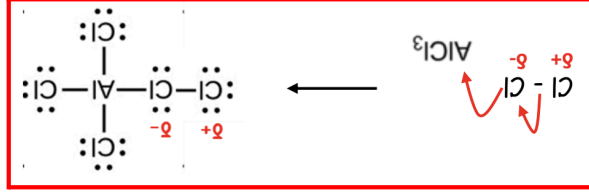
Halogenation of Benzene



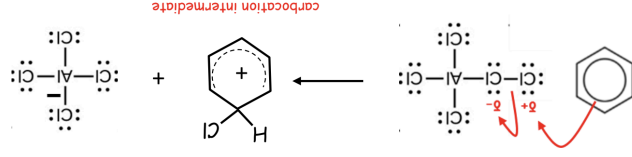
Conditions:

- Room temperature
- Absence of light
- Anhydrous conditions (prevents hydrolysis of halogen carrier)
- Presence of a halogen carrier e.g. AlCl_3 or FeCl_3

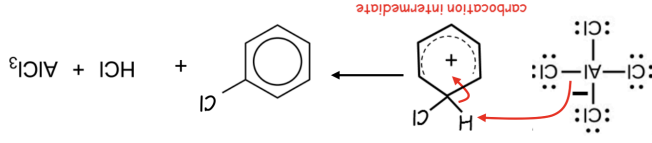
Step 1 – Generation of Electrophile



Step 2 – Attack of Electrophile on Benzene Ring

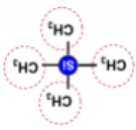


Step 3 – Abstraction of Proton & Regeneration to Catalyst

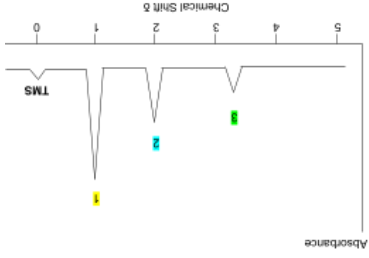
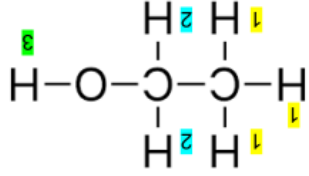
High Resolution ^1H NMR Spectra

Interpreting NMR Spectra

Standard Reference Compound – Tetramethylsilane (TMS) (Do not need to learn mechanism / why TMS is used)



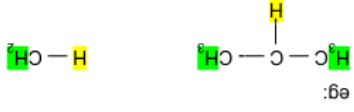
Ethanol (Low Resolution) (AS)



- Ethanol has 3 proton environment $\rightarrow$ 3 absorption peak
- Area under each peak is proportional to number of hydrogen in the environment
- Radiowave is absorbed

How does ^1H NMR work?

1. Identify number of Hydrogen environment (same as low resolution)
2. Count the number of Hydrogen on the adjacent carbon
3. $n+1$ rule $\rightarrow$ where n is the number of hydrogen on the adjacent carbon
4. $n+1$ shows number of splitting peaks on the nmr



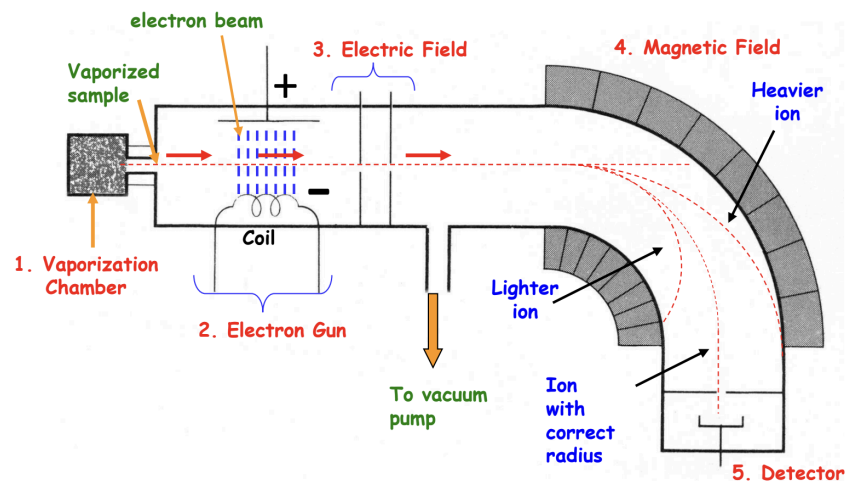
The H will split into $6+1$ (7) peaks due to 6 H on the adjacent carbon

The H will split into $2+1$ (3) peaks due to 2 H on the adjacent carbon

Note: usually WJEC maximum will split into 4–5 peaks (less complicated than you think)

4.8 Organic Analysis

Mass Spectrometry



Vaporised → Ionised → Acceleration → Deflection → Detection

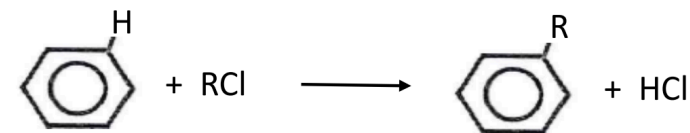
- Electrons collide with the gaseous atoms: $M_{(g)} + e^- \rightarrow M^+_{(g)} + 2e^-$
- Acceleration: Electric field
- Deflection: Magnetic field to separate different m/z

Common Fragments

CH_3^+	15
$CH_3CH_2^+$	29
$CH_3CH_2CH_2^+$	43
$COOH^+$	45
CO^+	28
$C_6H_5^+ / C_6H_5CH_2^+$	77 / 91

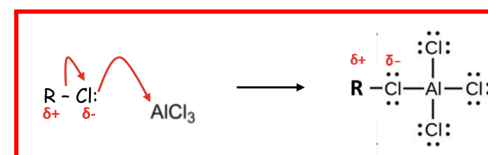
Alkylation

Overall Reaction:

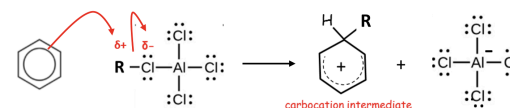


Conditions: Room temperature
 Anhydrous conditions (prevents hydrolysis of halogen carrier)
 Presence of a halogen carrier e.g. AlCl_3 or FeCl_3
 The appropriate halogenoalkane

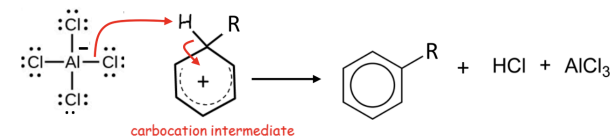
Step 1 – Generation of Electrophile



Step 2 – Attack of Electrophile on Benzene Ring



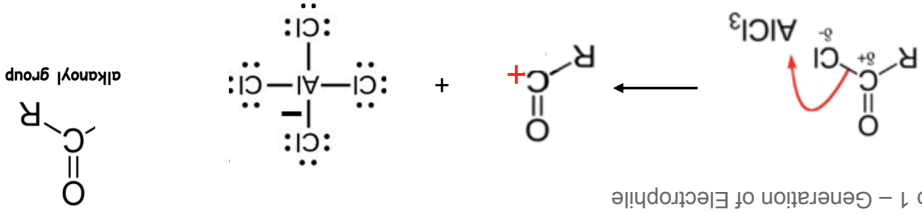
Step 3 – Abstraction of Proton & Regeneration of Catalyst



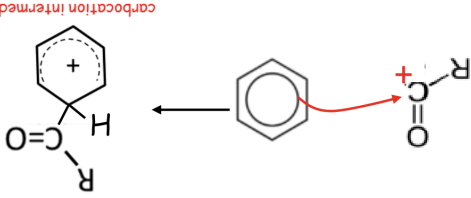
Acylation (Friedel-Crafts)

Conditions: Approx. 60°C for about 30 minutes
 Presence of $AlCl_3$ as the catalyst
 A mixture of an acyl chloride:
 e.g. ethanoyl chloride (CH_3COCl) and benzene

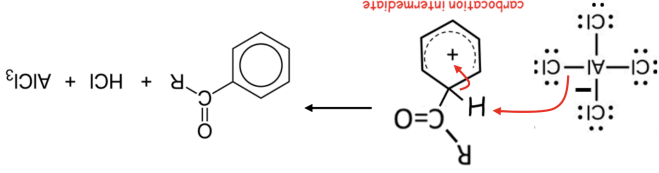
Step 1 – Generation of Electrophile



Step 2 – Attack of Electrophile on Benzene Ring



Step 3 – Abstraction of Proton & Regeneration to Catalyst



Organic Synthesis

Organic Synthesis Technique

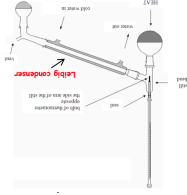
Recrystallization & Purification

Reflux

- Liquid evaporates, vapour goes into condenser, cools and condenses
- The reaction is slow so it allows time for equilibrium to be established
- Stops reactants / products boiling away

Simple Distillation

- Separate volatile liquid from dissolved non volatile impurities
- Liquid with the lowest bp will distil out

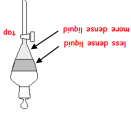


Buchner Filtration - Vacuum Filtration

- Separated an insoluble solid from a liquid

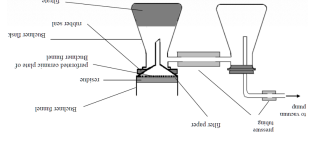
Separating Funnel

- Separate two immiscible liquids

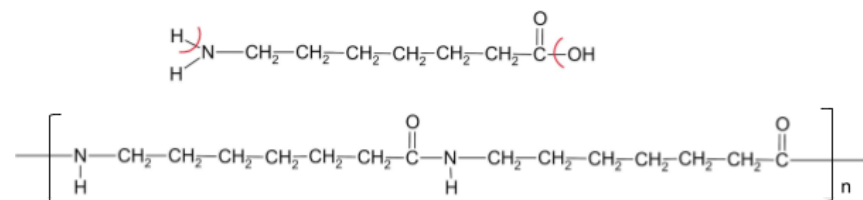


Steam Distillation

R_f = distance travelled by spot (Center of Spot)
 distance travelled by solvent front



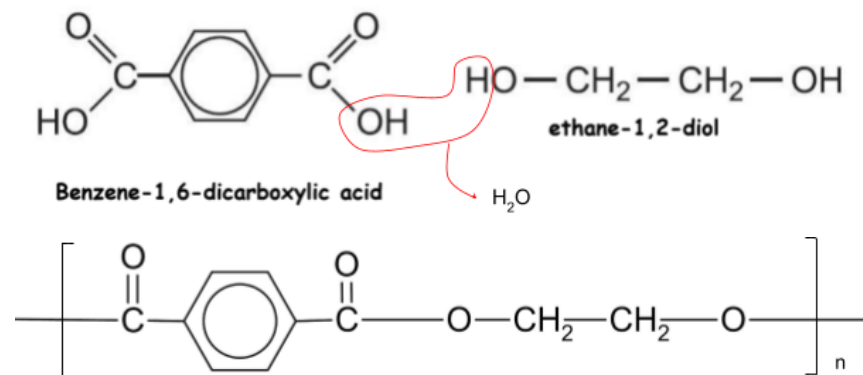
Polyamides (Nylon 6)



- Amine and Carboxylic Acid
- Hydrogen Bond are present
- High melting point, high tensile strength (physically strong), insoluble

Polyesters (Polyethyleneterephthalate)

- Diol and a Dicarboxylic acid

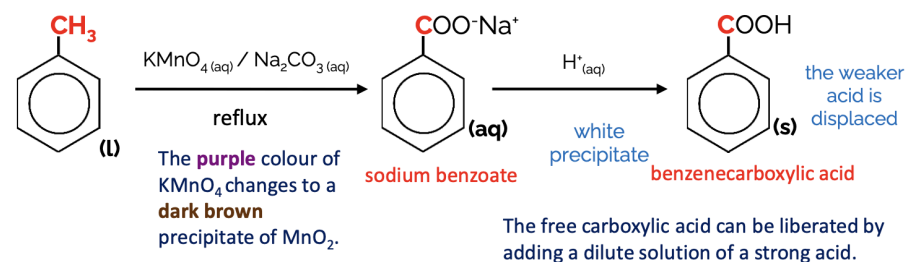


Condition:
 Sb_2O_3 catalyst;
 low pressure;
 260°C

- Permanent dipole-dipole and vdw only $\rightarrow$ less mechanical strength than nylons
- PET: strong, stiff synthetic fibres $\rightarrow$ manufacture of clothing
- Acids and alkalis will cause hydrolysis of the ester and amide linkages

Oxidation of Alkyl Side of Benzene Ring

Conditions: Reflux followed by acidification.
 Potassium manganate(VII) solution made *alkaline* with sodium carbonate, acts as an oxidising agent.



MnO_2 brown ppt (under alkaline medium / x acidic medium)

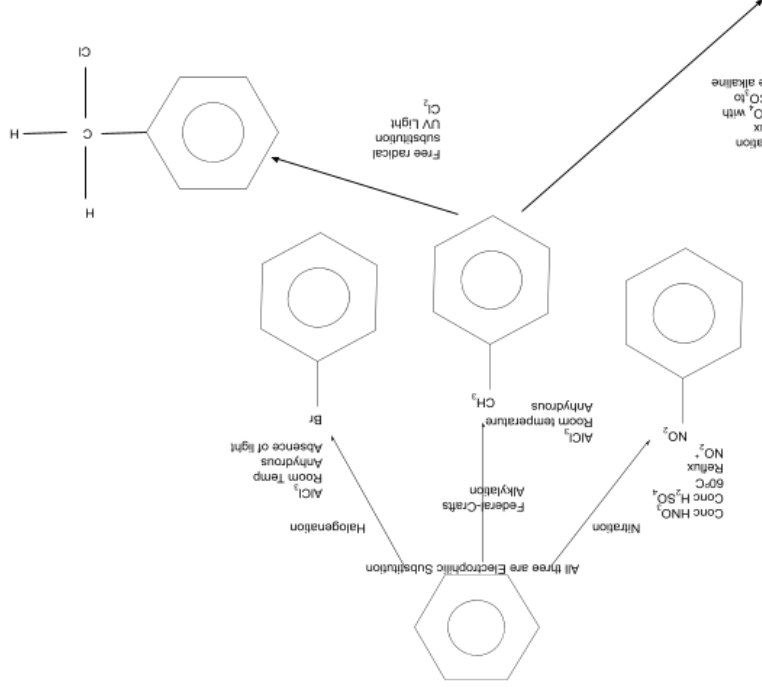
Comparison of the Reactions of 1-Chlorobutane and Chlorobenzene with $\text{OH}^-_{(\text{aq})}$

- Chlorobenzene is more difficult to Nucleophilic Substitution compared to 1-Chlorobutane

Because:

- Repels the attacking OH^- nucleophile away from carbon atom C_1 .
- Strengthens the $\text{C}-\text{Cl}$ bond giving the bond some π bonding character.

Summary reaction of benzene



Reminder Question Key Terms:
 Mechanism: *Free Radical Substitution, Electrophilic Addition / Substitution, SN*
 Type of reaction: *Nitration, Acylation, Alkylation*



Condensation Polymer

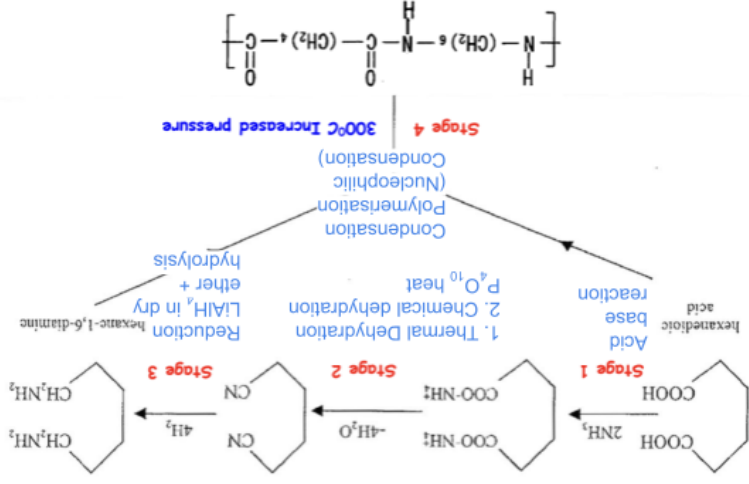
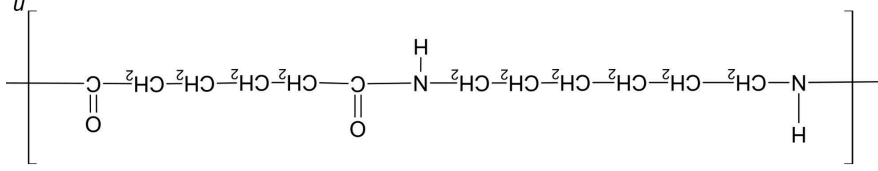
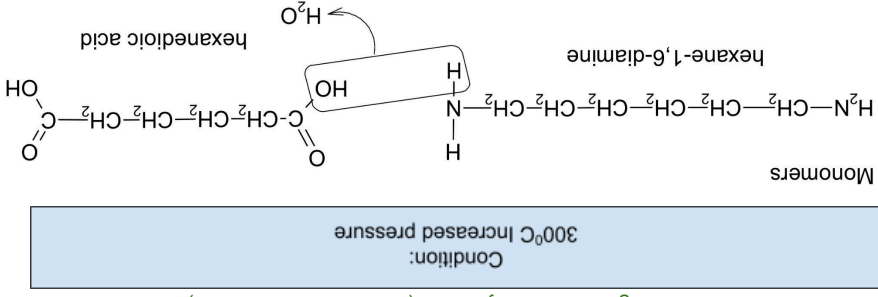
- Condensation: Elimination of small molecules
- Nylon and PET examples

Polyamides (Nylon 6:6)

- Diamine and a Dicarboxylic Acid
- Uses of Nylons: Textiles, fishing line, ropes, cables and carpets.

Number of carbon chain length in diamine (6 C in hexane-1,6-diamine)

Number of carbon chain length in dicarboxylic acid (6 C in hexanedioic acid)



4.8 Organic Synthesis

Compare and Contrast Addition and Condensation

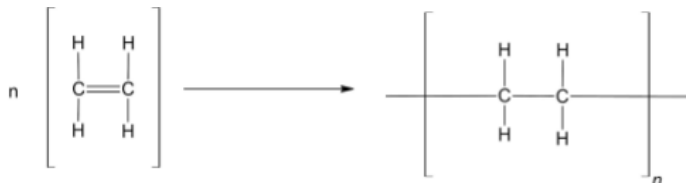
Similarities - Both can be represented by a repeating unit - Both involve large number of small molecules unit joining together to form a long chain molecule	
Differences Addition - Monomers join together to form a single long chain polymers - Monomers has 1 functional group	Differences Condensation - Monomers join together to form a log chain polymers in elimination of small molecules - Monomers has 2 functional group

Addition Polymerisation (AS)

- Must have double bond for electrophilic addition reaction

Ethene:

- High temperature (250°C)
- High pressure (2000 atm)
- Mechanism: Free radical addition



4.3 Alcohol

Industrial Preparation of Alcohol from Alkenes

Conditions: 300°C
60-70atm
phosphoric (V) acid on a silica gel (catalyst)

The mechanism is not required.

Other Method of Alkene to Alcohol:

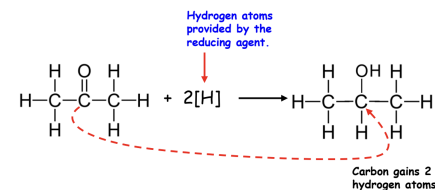
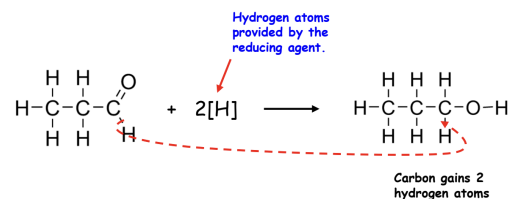
- Electrophilic Addition (and Hydrolysis) to form Alcohol

Nucleophilic Substitution from Halogenoalkane to Alcohol

Conditions: aqueous NaOH
reflux
mutual solvent (e.g. propanone)

Reduction from Carbonyl Compound (Aldehyde & Ketone) – Nucleophilic Addition

Conditions: aqueous NaBH₄ (sodium borohydride/tetrahydridoborate (III))
room temperature
separation of alcohol by fractional distillation



Reduction from Carboxylic Acid

Conditions: LiAlH₄ (lithium aluminium hydride/tetrahydridoaluminate (III)) in a dry ether solvent
room temperature
sulfuric acid

Fermentation from Sugar

Conditions: yeast ("zymase")
no air (anaerobic conditions)
optimum temperature and pH for the yeast
separation of ethanol by fractional distillation

Fermentation of glucose			
Advantages	- Renewable material - Low cost	Disadvantages	- Low atom economy - Low yield - Greenhouse gas emission - Purification and separation are expensive
Direct hydration of ethene		Advantages	- More pure product → Higher yield - High atom economy - No toxic / harmful product
Disadvantages	- Non renewable raw material (crude oil) - High pressure requires → Can be dangerous + expensive		

Structure of Protein

Primary Structure

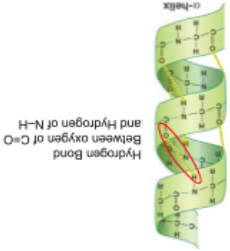
Primary structure is the sequence and number of amino acids in a polypeptide.



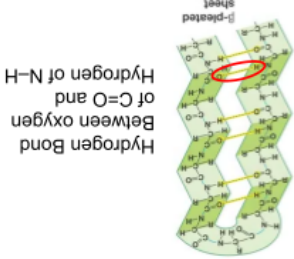
Secondary Structure

Secondary structure of a protein is the coiling or folding of the polypeptide into an alpha helix and beta pleated sheet with hydrogen bonds.

Alpha Helix



Beta Pleated Sheet



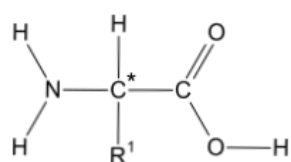
Function of Protein Notes

- Protein provide structure and strength to cells eg: collagen
- Muscle contraction involves Actin & Myosin
- Biological Catalyst
- Amylase (enzyme) catalyst hydrolysis of starch into glucose
- Proteins regulate hormones and cell division
- Protein and the immune system

Tertiary structure is the folding and coiling of the secondary structure to a complex 3D shape with ionic, disulfide, hydrophobic interaction and further hydrogen bond.

Tertiary Structure

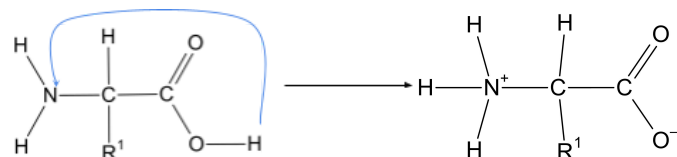
4.7 Amino Acid



Properties

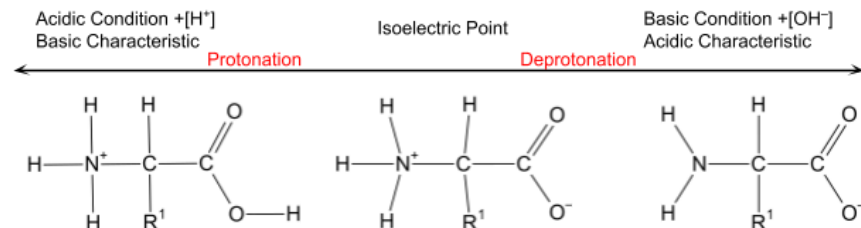
- White crystal at room temperature
- Insoluble in non polar solvent
- Soluble in water

Zwitterion

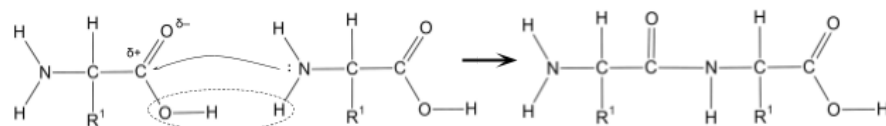


Amphoteric Character

Amino acids show both acidic and basic character reacting with both acids and bases



Formation of Dipeptides – Nucleophilic Condensation



Condensation: Elimination of water

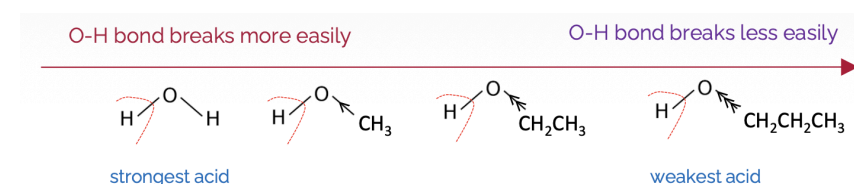
Chemical Properties of Alcohol

Alcohol Acidic Character

- Alcohol react slowly with **sodium** to evolve hydrogen gas
- Alcohol react in a similar way to water, i.e. alcohol have some acidic character
- Alcohol are weaker acid than water

Observation:

- Colourless gas forms (H_2 Gas)
- White solid forms $\text{CH}_3\text{CH}_2\text{O}^-\text{Na}^+$



- Longer the carbon chain, the inductive effect (pushing the electron) is stronger
- The O-H bond is strengthened if electron being donated to O-H
- *Easier to break O-H bond, stronger acidic character (as it can donate H^+)*

Summary

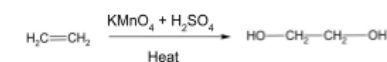
- Alkyl groups have an electron donating (pushing) inductive effect. This increases the electron density in the O-H bond.
- The bond is therefore strengthened when compared to that in water and breaks less easily. This makes all alcohols weaker acids than water.
- As the length of the carbon chain increases, the inductive effect increases in magnitude. This further strengthens the O-H bond and reduces the acid character.

Elimination of Halogenoalkane to Ethene

- Alcoholic sodium hydroxide solution
- HBr / H-X is removed

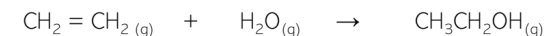
Oxidation of Ethene to Diol (2 OH group Alcohol)

- $\text{KMnO}_4 + \text{H}_2\text{SO}_4$
- Heat



Industrial Preparation of Alcohol (from alkanes)

- Phosphoric acid on silica gel catalyst
- 300°C
- 60-70 atm



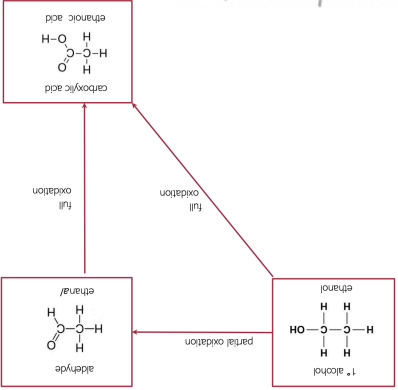
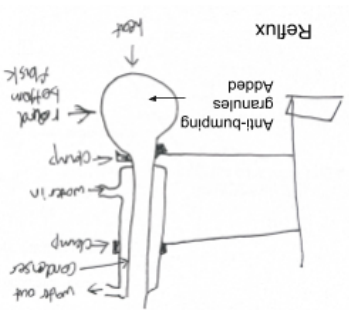
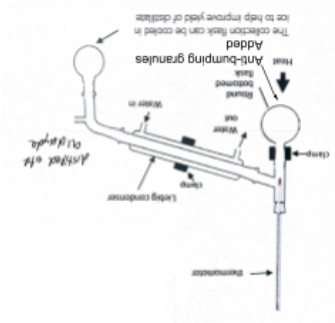
Oxidation of Alcohols (AS)

Condition

- Acidified Potassium Dichromate
- Concentrated Sulphuric Acid (Catalyst)

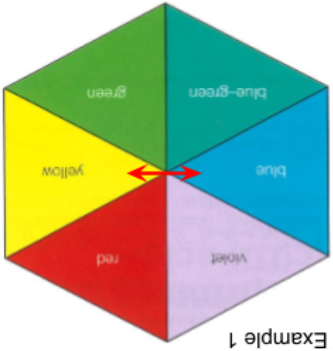
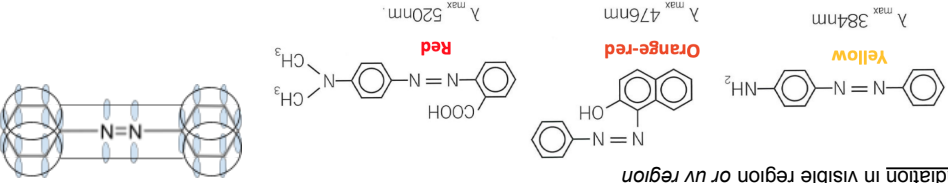
Primary Alcohol

1. Partial Oxidation
 - Distill out the aldehyde as it forms to prevent further oxidation to carboxylic acid
 - Excess alcohol
 - Full oxidation
 - Reflux
 - Excess oxidising agent – to ensure maximum yield of carboxylic acid



Role of -N=N- Chromophore in Azo Dyes

Chromophore – Structural unit in a molecule which is responsible for the absorption of light radiation in visible region or uv region



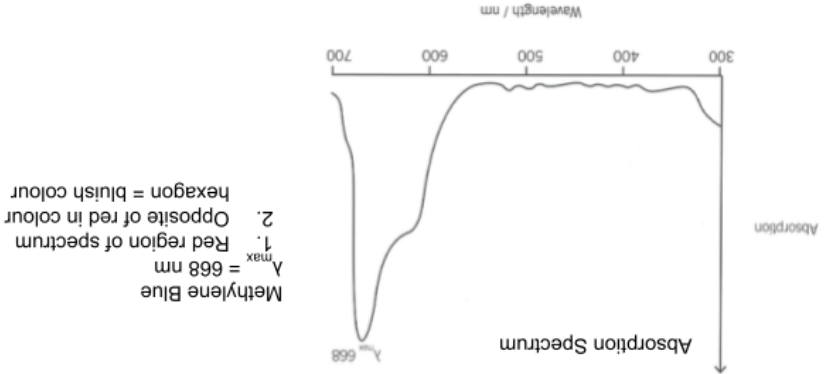
Example 2: Relationship between color and absorption spectrum.

1. λ_{max} is blue

2. Absorption spectrum = opposite colour of hexagon

Colour	Wavelength/nm
Red	620-750
Orange	580-620
Yellow	560-580
Green	490-560
Blue	430-490
Violet	380-430

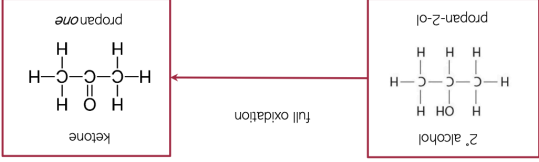
Example 2 Methylene Blue visible spectrum



Give the reason of the dye is a certain colour in white light:
The rest of visible spectrum is absorbed

Tertiary Alcohol

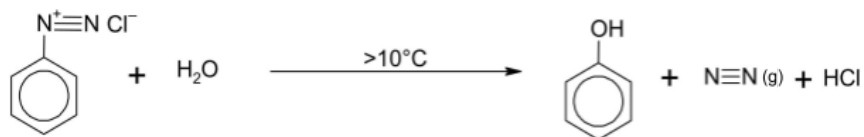
- NO Reaction
- As it would require breaking of C-C bond



Secondary Alcohol

- Full Oxidation
- Acidified Potassium Dichromate
- Concentrated Sulphuric Acid
- Reflux

If reaction is above 10°C, diazonium salt decompose ← unstable

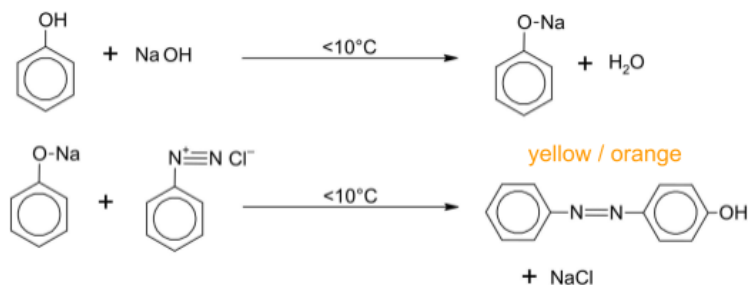


Bubbles of N₂ are seen again

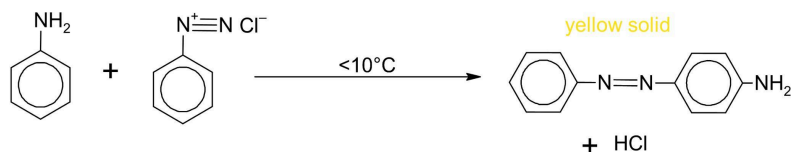
Coupling Reactions

Diazonium salt react with aromatic compound to form **azo dyes** by electrophilic substitution

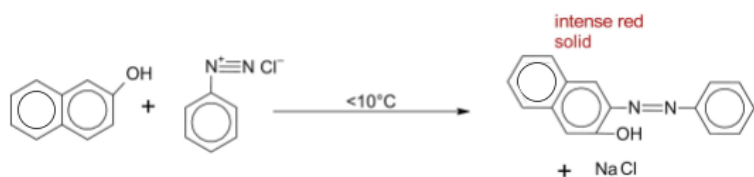
Phenol



Phnylamine



Napthalen-2-ol



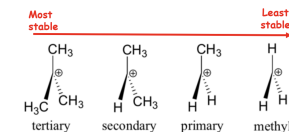
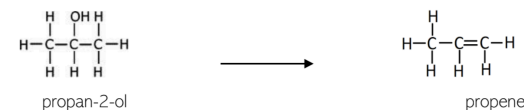
Azo Dyes – Two benzene ring linked by N=N

Azo compound – Highly delocalised system → Delocalise electron jump to high energy level

Dehydration Alcohol to Alkenes (Elimination) (AS)

Condition

- Concentrated H₂SO₄ (dehydration agent + catalyst)
- Heat to 180°C

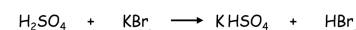


Conversion of Alcohol to Halogenoalkane

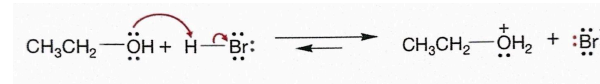
- Alcohols do not undergo nucleophilic substitution with Cl⁻ or Br⁻ ions alone.
- OH⁻ group of the alcohol is highly basic and is a very poor leaving group

Bromination

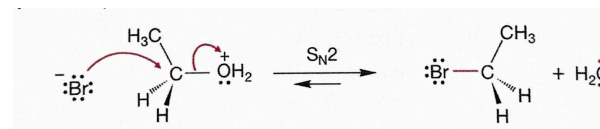
Step 1 – HBr Generated “in situ” from H₂SO₄ and KBr



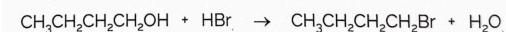
Step 2 – Propagation of OH group



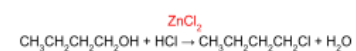
Step 3 – Nucleophilic Attack



Overall Reaction



Chlorination



Conc HCl and ZnCl₂ act as catalyst

1. The alcohol is shaken in a stoppered flask with concentrated hydrochloric acid and

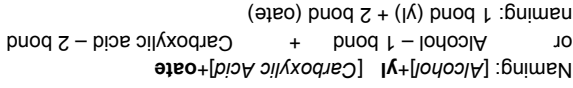
- The halogenoalkane formed as the top immiscible layer it is removed using a separating funnel
- Excess sodium carbonate solution is added to the halogenoalkane to neutralise any remaining acid and shake. The stopper needs to be removed at regular intervals.
- The halogenoalkane is again separated from the water layer using a separating



- ### Conversion to Esters (Carboxylic Acid → Esters)

H₂SO₄ catalyst (catalyst)

- Reaction is reversible so percentage yield might not be 100%



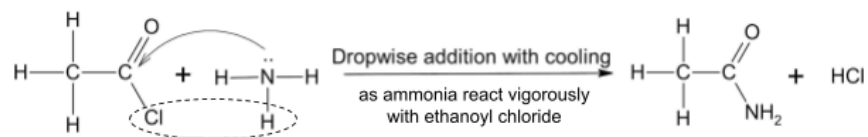
Step 1: Preparation of Nitrous Acid

$${}^{(aq)}\text{IO}_3^- + {}^{(aq)}\text{HNO}_2 \leftarrow {}^{(aq)}\text{HCl} + {}^{(aq)}\text{NO}_2^-$$
$$\text{H}_3\text{C}-\text{CH}_2\text{CH}_2\text{CH}_2-\text{N}\equiv\text{N}^+\text{Cl}^- + \text{H}_2\text{O} \longrightarrow \text{H}_3\text{C}-\text{CH}_2\text{CH}_2\text{CH}_2-\text{OH} + \text{HCl} + \text{N}_2$$
$$\text{Ph-NH}_2 \xleftarrow[\text{HNO}_2]{>10^\circ\text{C}} \text{Ph-N}^+\equiv\text{N Cl}^-$$

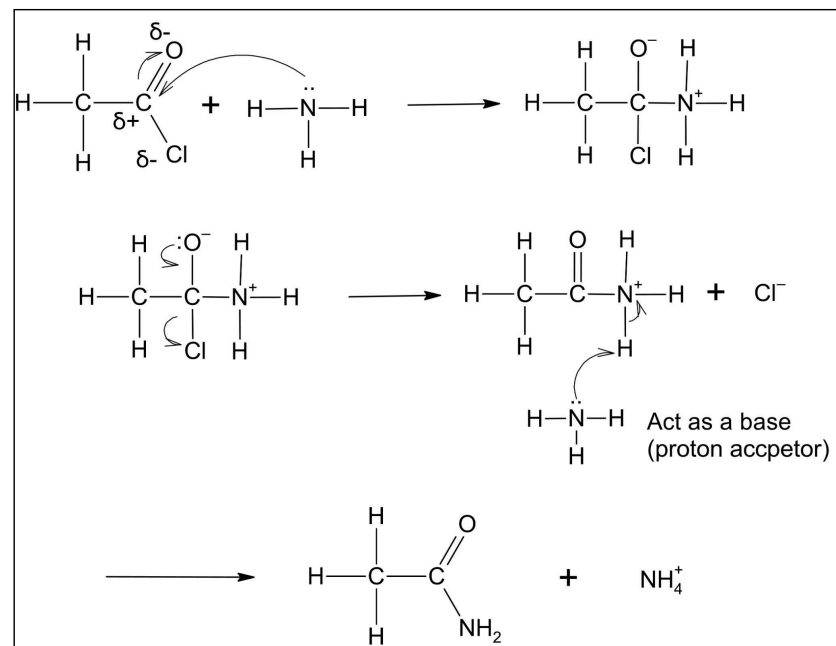
Test for Aliphatic and Aromatic Amine:
No bubbles of N_2 gas in the aromatic amine in the step above.

Reaction as Nucleophiles

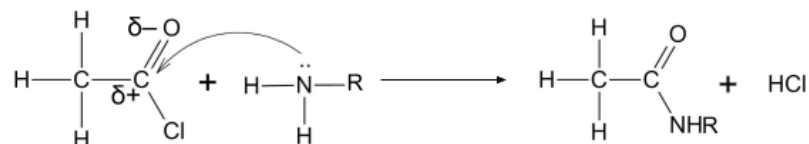
Ethanoylation (1° Amine / Ammonia and Ethanoyl Chloride)



Nucleophilic Addition Elimination Mechanism (Not Required for WJEC)

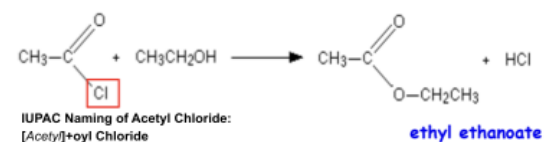


Similarly amines (both aromatic and aliphatic) would react similarly to form N-substituted amides



- HCl produce can react with excess amine to form amine salt
- Amides have sharp melting points
- Excess Ammonia (or NH_2R) will form amine salt with HCl which is a product)

Conversion to Ester from Acyl Chlorides – Nucleophilic Addition Elimination

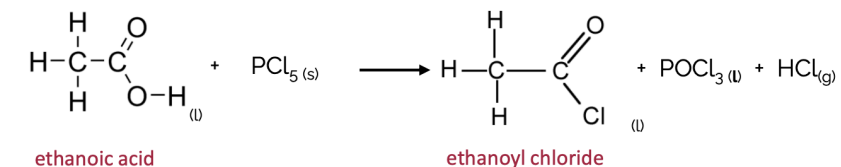


Conditions:

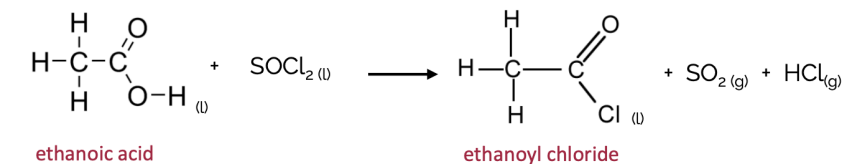
- Acyl chloride
 - Ice bath
 - Fractional distillation (to separate the ester from the reaction mixture)
- Ethanoyl chloride reacts instantly with ethanol
 - It is very exothermic reaction and HCl (steamy acidic gas is given off)
 - The alcohol must be added dropwise with cooling → reduce the vigour of the reaction

Preparation of acyl chloride

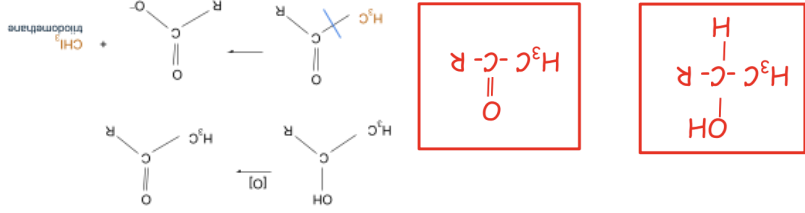
1. PCl_5



2. SOCl_2



Iodoform Reaction



Physical Properties of Alcohol

Melting and Boiling Points

MP BP increase as the carbon length increase

Comparing Butane, Chloroethane and Propan-1-ol

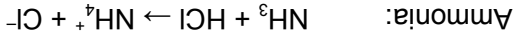
Butane – Non polar (only weak vdw)
 Chloroethane – C–Cl bond is polar (permanent dipole)
 Propan-1-ol – H and O lone pair can form Hydrogen bond

Solubility in Water

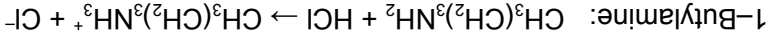
Shorter carbon chain alcohol are soluble in water
 As they can form hydrogen bond with water

Longer carbon chain: solubility decrease
 As the chain length increase, a larger proportion of the molecule becomes non-polar and cannot form hydrogen bond with water

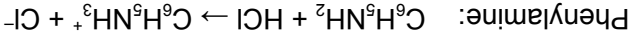
Amine Reaction with Acid – Salt Formation



Observation:
 Smell of ammonia disappears as an involatile ionic salt is formed.

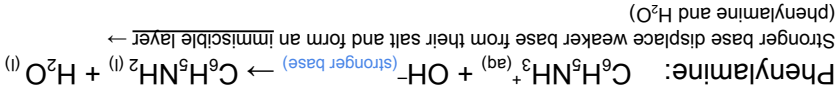


Observation:
 Smell of fish disappears as an involatile ionic salt is formed.
 Liquid amine dissolves into solution.



Observation:
 Smell of fish disappears as an involatile ionic salt is formed.
 Liquid amine dissolves into solution.

Base Displacement (reverse)



Physical and Chemical Properties of Amines

Melting and Boiling Points

- Can form H bond → Higher mp & bp
- Oxygen more electronegative than Nitrogen → Alcohol > Amine mp & bp

Solubility in Water

- Aliphatic amine: increasing length of alkyl side chain reduces solubility
- Aromatic amine: slightly soluble in water (lone pair delocalised into ring so lone pair is less available for hydrogen bond to water)

Basic Character Strength

Compare: The availability of lone pair on nitrogen atom act as proton acceptors (bases)

Ammonia



1-Propylamine



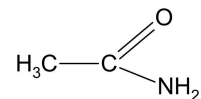
There is a stronger inductive effect on 1-propylamine, so the lone pair on nitrogen on 1-propylamine is more available for reaction
→ 1-propylamine (is a stronger base) > ammonia

Phenylamine



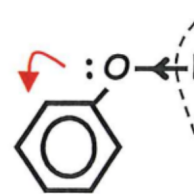
Lone pair on nitrogen delocalise into benzene ring → Less available to pick up a proton

Ethanamide

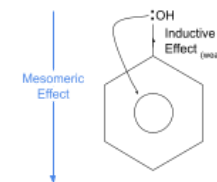


Conjugation of nitrogen lone pair with the carbonyl system
→ so lone pair on nitrogen is less available to form coordinate bond with proton

4.3 Phenol



The lone pair of electrons on the oxygen atom is delocalised into the pi system of the ring. → Increasing electron density on ring
This weakens the O-H bond.

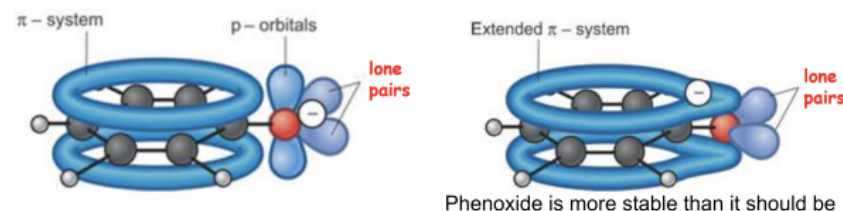


Mesomeric Effect

- Electronegative oxygen atom draws the electron towards itself in C-O bond (inductive)
- A lone pair of electrons on oxygen delocalised into the ring (mesomeric effect)
- Mesomeric Effect > Inductive Effect

Acidic Character (O-H bond strength so H⁺ can be released)

- Phenol is a stronger acid compared to alcohol
 - The delocalisation of the lone pair into the ring, reduces electron density in O-H bond, this **weakens the O-H bond**
 - The phenolate ion is stabilised by the delocalisation of the negative charge into the ring (shown in image below) [the p orbital interact with the ring]



Phenoxide is more stable than it should be

Solubility in Water

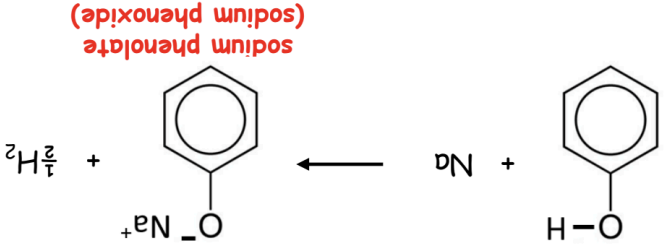
Phenol is moderately soluble in water

One pair of the lone pair is inside of the ring, so less chance of forming H-bond with water

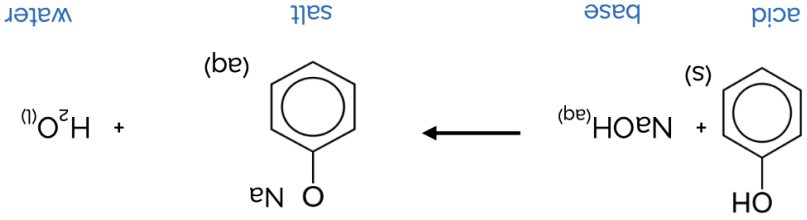
Reaction of Phenol

With Sodium Metal

Molten phenol release a proton when sodium is added

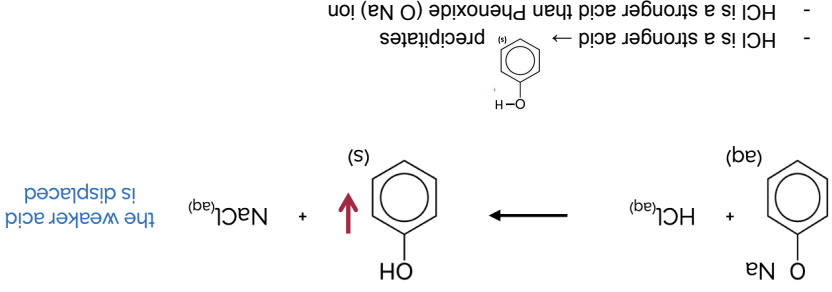


With NaOH



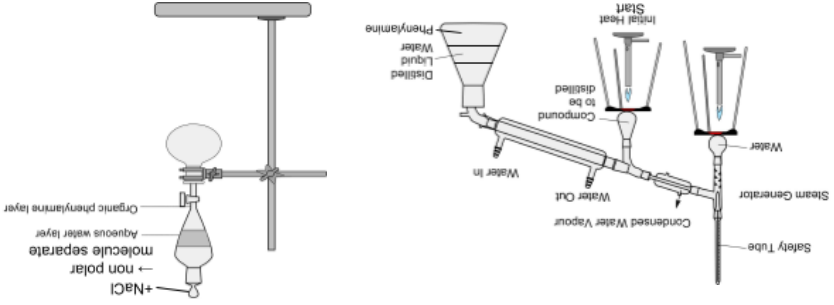
When NaOH is added, the phenol readily dissolves into solution forming ionic salt
This confirms that phenol has acidic character

Followed by addition of HCl

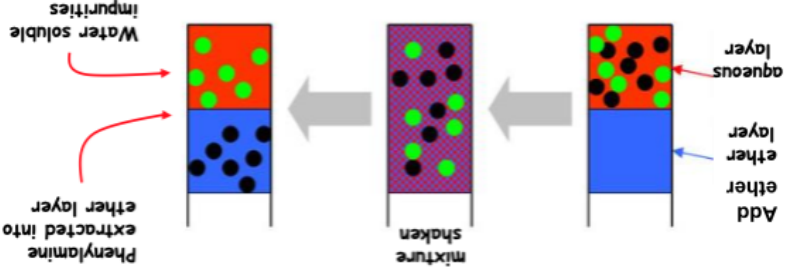


Steam Distillation of Phenylamine

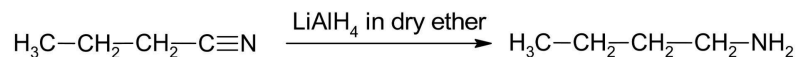
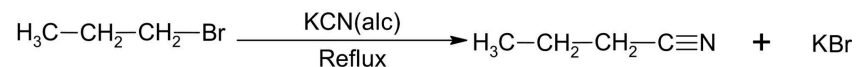
- Separating 2 immiscible liquid



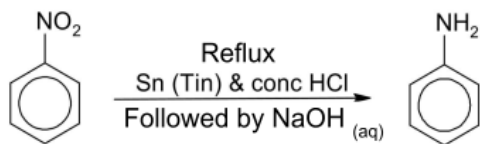
1. Steam generated by heating water in another flask is blown through the mixture and the water and phenylamine turn to vapour.
2. As the hot steam passes through the mixture it condenses, releasing heat. This will be enough to boil the mixture of water and phenylamine.
3. The vapour is then condensed and collected.
4. **Salting Out:** Solid sodium chloride is added to the distillate in a separating funnel and the mixture shaken. This 'forces' most of the phenylamine or non-polar organic molecules out of the aqueous solution.
5. The organic layer of phenylamine is denser than water and so the lower layer is carefully run off into a clean beaker.
6. **Solvent Extraction:** The upper aqueous layer will still contain some tiny amounts of phenylamine is more soluble, like ethoxyethane ether. This can be extracted into another solvent in which the phenylamine layer has been run off, the ether is added to the aqueous layer in the separating funnel. The funnel is stopped and shaken. Most of the phenylamine passes into the ether layer.
8. The ether layer is then run off and collected into a clean beaker. This solvent is highly volatile and so can then be allowed to evaporate
9. Finally distill to collect the phenylamine



From Halogenoalkane to Amine (step 2 same as above)

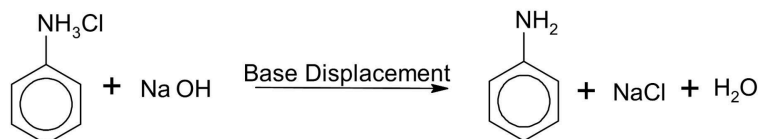
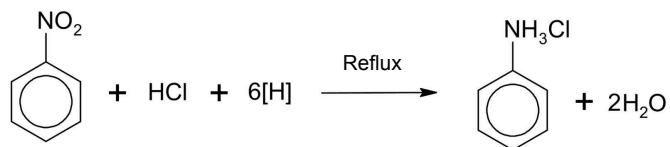


Preparation of Aromatic Amine

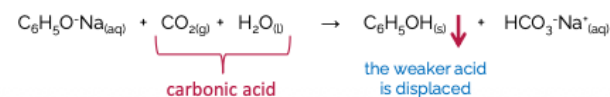


- Tin is not a catalyst
- Reducing agent not LiAlH_4 – prevent breaking of benzene ring

Detail of the 2 step reaction:



Followed by addition of CO_2



- Shows carbonic acid is stronger acid than phenol

Followed by Sodium Carbonate or Sodium Hydrogencarbonate

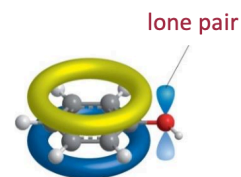
- Na_2CO_3 and NaHCO_3 are not a strong enough acid than phenolate

Esterification

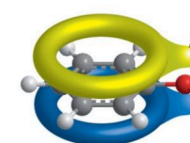
- **Phenol will not react directly with carboxylic acid to form esters**

Because:

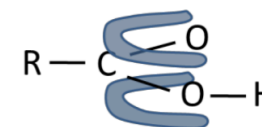
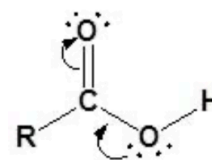
- Phenol - the lone pair on the oxygen of phenol is delocalised into the benzene ring thereby reducing its availability for nucleophilic attack



extended delocalised system

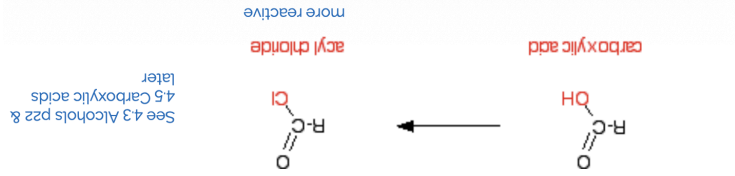


- Carboxylic acid - conjugation of the lone pair on oxygen in the $-\text{OH}$ group with the π bond of the adjacent $-\text{C}=\text{O}$ group reduces the electropositive character of the C in the $-\text{C}=\text{O}$ group. This means a stronger nucleophile than phenol is required.

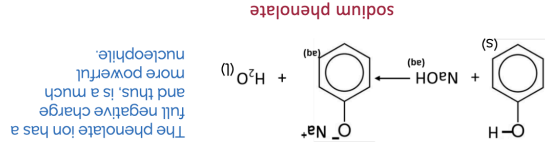


Esters from phenol can only be made from acyl chloride

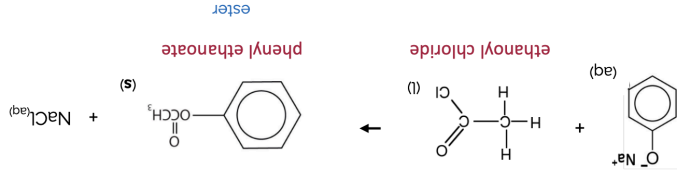
- i. Convert the carboxylic acid into an acyl chloride using either SOCl_2 or PCl_5 .



- ii. Dissolve phenol in sodium hydroxide solution to create the phenolate ion.



- iii. Shake the sodium phenolate with the acyl chloride and the reaction will occur at room temperature to produce the ester as a white solid in suspension.

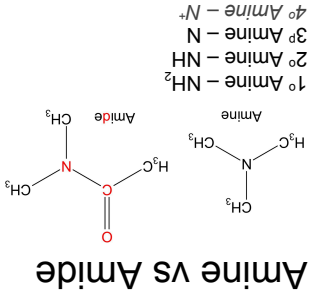
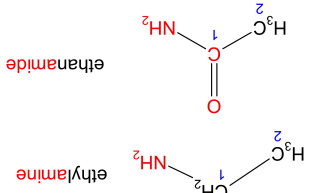


Chemical Properties compared with Benzene

- Both react with electrophiles and undergo substitution reactions
- In phenol delocalisation of the oxygen electron pair into the ring, increases the electron density in the π bonding system $\rightarrow$ electrophiles will attack phenol far more readily than benzene

4.6 Amines

Nomenclature



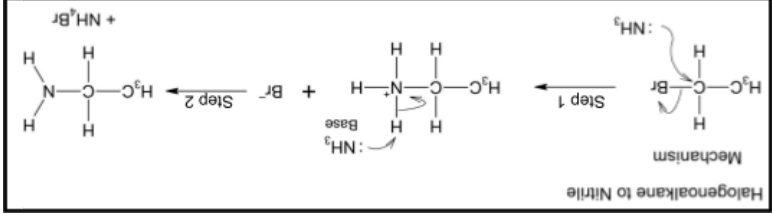
Preparation of Aliphatic Amine

From Halogenoalkane

Reaction Equation



- Excess Concentrated Alcoholic (ethanol) Ammonia solution
- Heat in a sealed container



As step 2 is a reversible equilibria, it requires a strong base (NaOH to displace):
 $\text{CH}_3\text{CH}_2\text{NH}_3^+\text{Br}^- + \text{NaOH} \rightarrow \text{CH}_3\text{CH}_2\text{NH}_2 + \text{NaBr} + \text{H}_2\text{O}$

From Nitrile (Covered in Carboxylic Acid Notes)

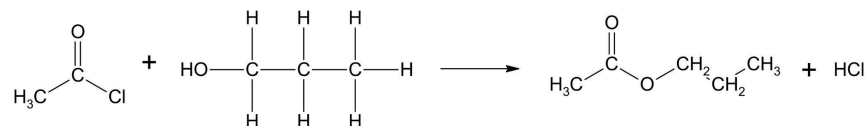
Reduction of Nitrile



- LiAlH_4 in dry ether and Hydrolysis of Water

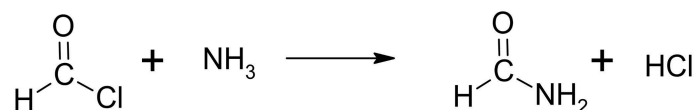
Further Reaction of Acyl Chloride

Acyl Chloride with Alcohol – Form Ester

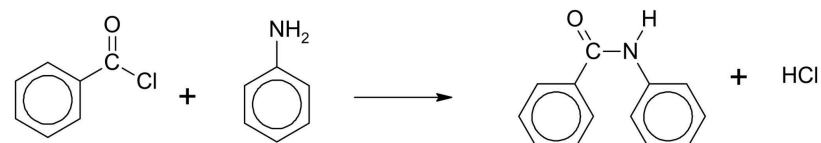


- Reaction is more vigorous than carboxylic acid
- HCl is given off

Acyl Chloride with Ammonia – Form Amide

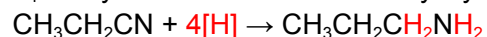


Acyl Chloride with Amine – Form N-Substituted Amide

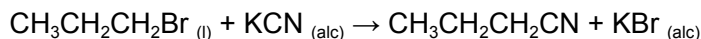


Reduction of Nitrile

Regagent: LiAlH_4 in dry ether solvent followed by Hydrolysis

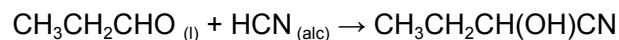


Synthesising Nitrile – From Halogenoalkane to Nitrile (Nucleophilic Substitution)



Increase length of chain (addition of 1 Carbon) + Reflux

Synthesising Hydroxynitrile (Nucleophilic Addition)



- Also could be done with KCN and H_2SO_4 (also +1 Carbon)

Purification of the Product by Recrystallisation (WJEC Mark Scheme)

1. Dissolve in minimum volume of hot solvent (water) in a hot water bath
 2. Filter insoluble impurities form
 3. Allow to cool for recrystallisation and pure solid forms
 4. Filter crystal + wash with cold solvent + dry with filter paper
- Eduqas 2017: Warm oven*

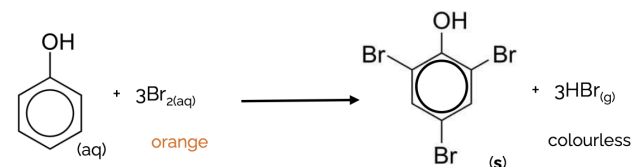
- The purified crystal can be checked by comparing it to the melting point

Reasons for Hot Minimal Solvent

Hot: Filter out insoluble impurities

Minimal: Maximum amount of precipitate can be crystallised out on cooling

Bromination of Phenol



Conditions:

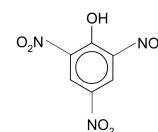
Bromine water
Room temperature
(no catalyst required)



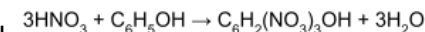
- White precipitate

- Colour change from yellow-orange to colourless
- Test for phenol

Reaction with nitric(V) acid solution (HNO_3)



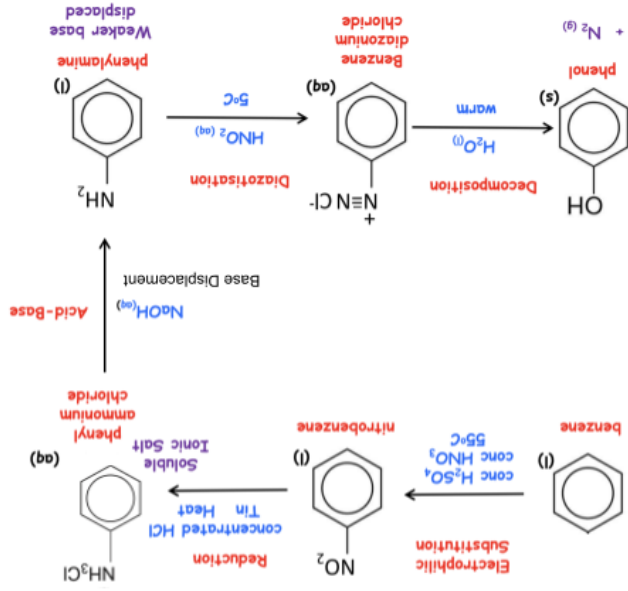
2,4,6-trinitrophenol



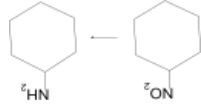
Conditions:

- Concentrated HNO_3 with dilute HNO_3
- No catalyst
- Room Temperature

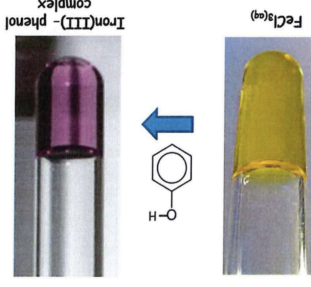
Preparation of Phenol in Lab

NO₂ to NH₂ (Reduction)

- Tin (reducing agent) and concentrated HCl
- Followed by NaOH + Reflux



Test for Phenol

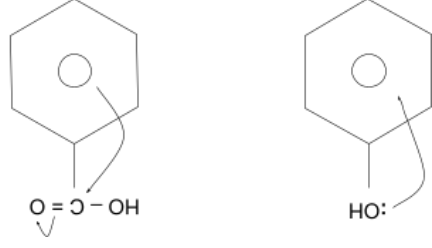


Phenol

Phenol – Has delocalise lone pair on oxygen, increase electron density on ring

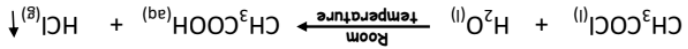
Benzene – Delocalised pi electron bonding system

Benzoic Acid – Conjugation of C=O pi electron with the pi system on the ring reduce electron density

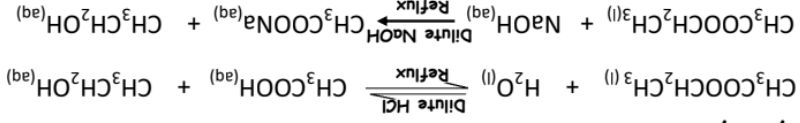


Overall Hydrolysis Equations

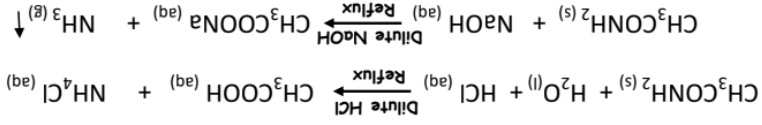
a. Hydrolysis Of Acid (Acyl) Chlorides



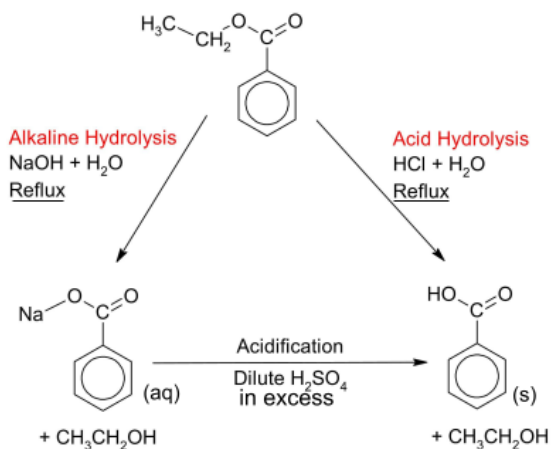
b. Hydrolysis Of Esters



c. Hydrolysis Of Amides

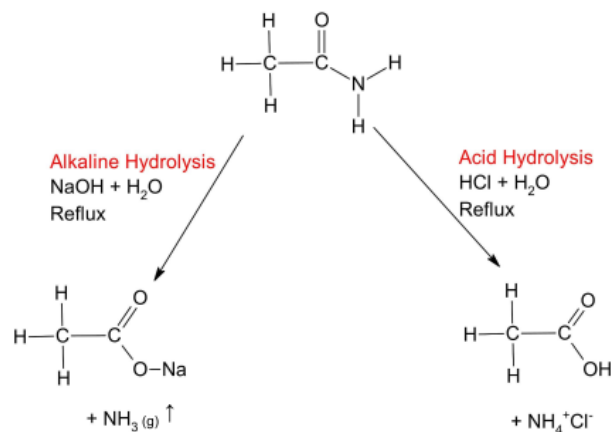


Ester Hydrolysis to C.A.



- Step 1: NaOH + H₂O, Reflux
- Step 2: Acidification: Dilute H₂SO₄
- To produce the carboxylic acid in alkaline hydrolysis, a dilute acid is added to hydrolyse the salt that was formed.
- Reflux → Allows reactant to completely react so solution does not have reactant
- Excess dilute H₂SO₄ / HCl use in acidification (preferably HNO₃ as it is more inert HCl is reactive)
- Use excess to maximise yield: Add excess HCl dropwise, until no white ppt forms

Amide Hydrolysis

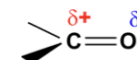


- Amide hydrolysis is difficult → Require reflux and Dilute acid / alkaline catalyst

4.4 Aldehyde and Ketone

Preparation of Carbonyl Compounds – Oxidation – AS

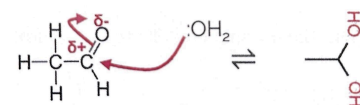
Polarity of Carbonyl Group



- Double bond in alkene has no polarity → Alkene are non-polar
- C=O bond is highly polarised → Aldehyde and Ketone are polar molecules

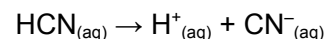
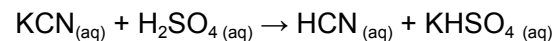
Physical Properties

- Aldehyde and Ketone have higher melting point compared to alkene
- Carbonyl compound have better solubility as their ability to form H-bond with water and forming a diol (which can form more H bond)
- Solubility decrease as the carbon chain increase



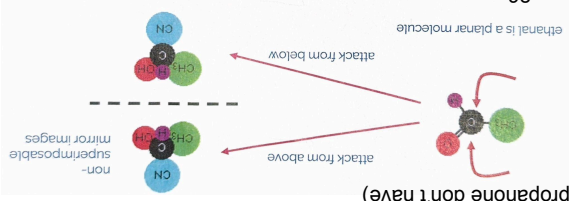
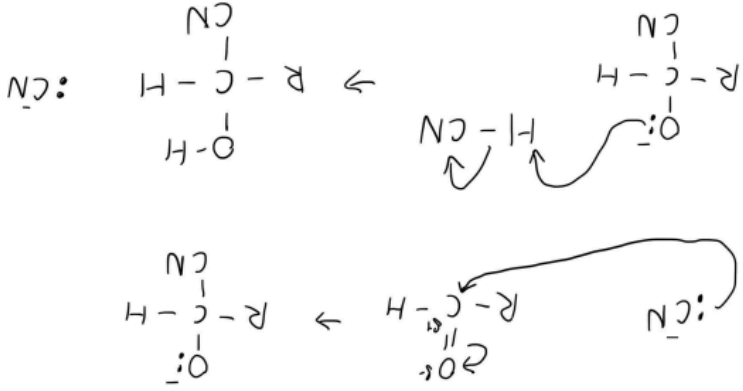
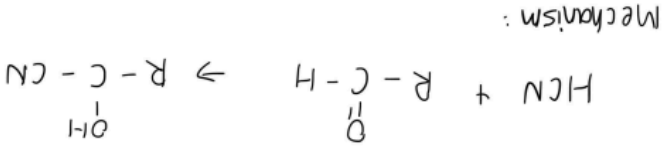
Step 1 – Generation of HCN

Reagent: KCN_(aq) and H₂SO₄ to form HCN (in situ)



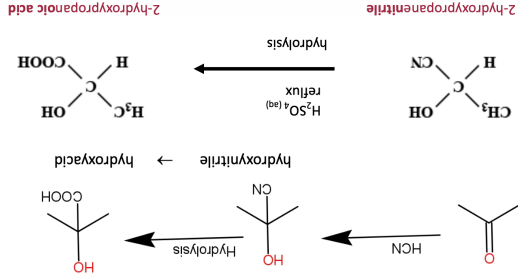
Step 2 – Nucleophilic Addition

1. Aldehyde and HCN



- Aldehyde may have optical isomer (methanal don't have)
- Ketone may have optical isomer (propanone don't have)

Notes:



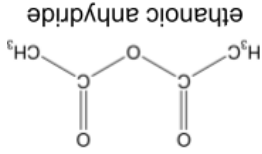
Hydrolysis of Acid Derivatives

Reactivity: Acyl Chloride > Ester / Amide

Acyl chloride hydrolysis to C.A.

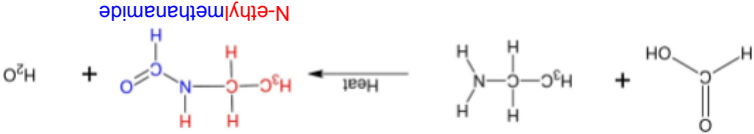


- Chlorine atom is highly electronegative (strong inductive effect)
- H-Cl (misty fumes)



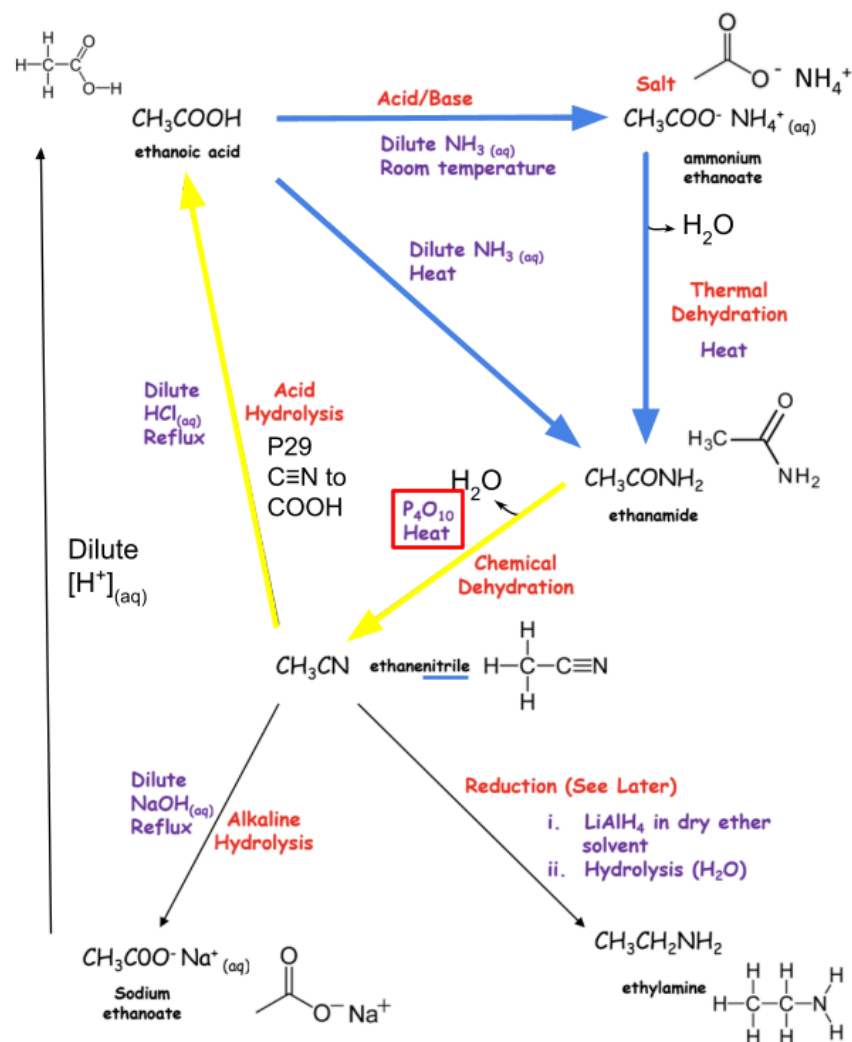
Carboxylic Acid to Acid Anhydrides (Out of Syllabus)

- Amines react on heating with ethanoic acid to form N-substituted amides. The amide forms as a white solid



Carboxylic Acid to N-Substituted Group

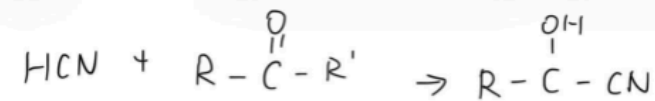
Conversion to Acid Amides and Nitriles



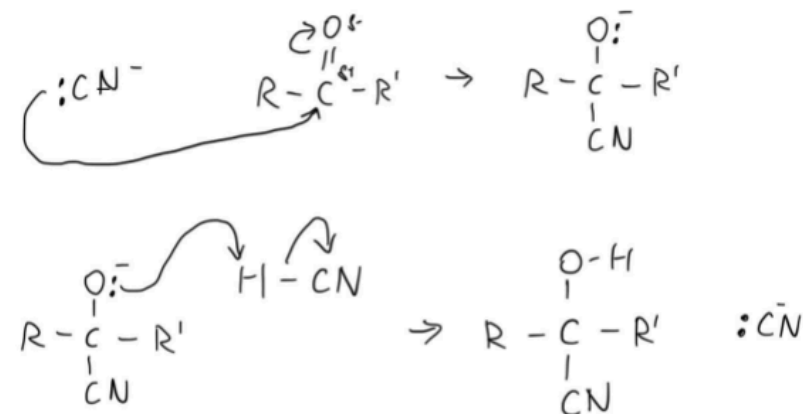
Amide $\rightarrow$ Nitrile by dehydration

- Heat
- P_4O_{10} (dehydrating agent)

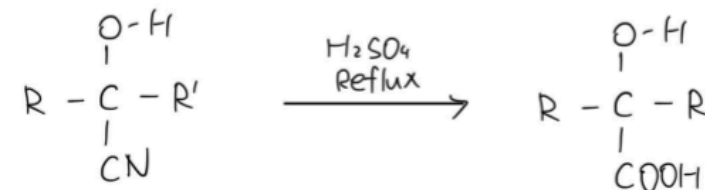
2. Ketone with HCN (identical with aldehyde)



Mechanism:



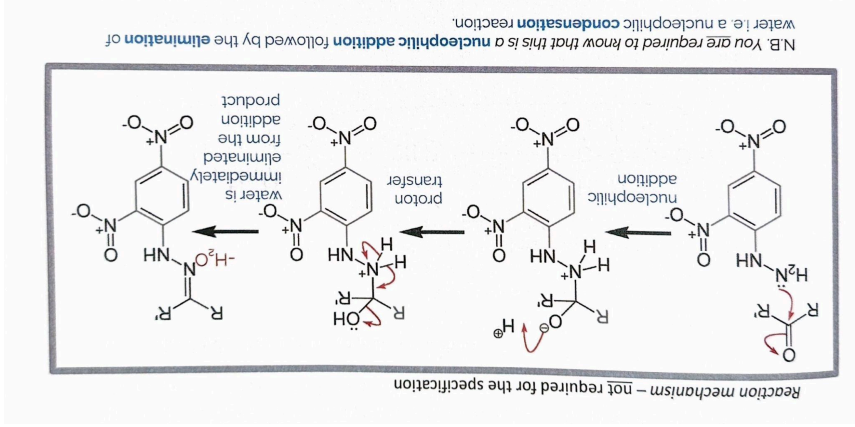
CN to Carboxylic Acid



Test for Carbonyl Compound – 3 Tests

2,4-DNPH (Brady's Reagent)

Reagent: 2,4-dinitrophenylhydrazine (2,4-DNPH) in alcohol solvent
 Observation: immediate formation of a heavy yellow orange precipitate
 Test for present of Aldehyde or Ketone



Identifying the Exact Compound

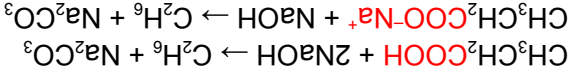
- Add a few drops of aldehyde or ketone into Brady's reagent (2,4-DNPH)
- Recrystallise
- A bright orange or yellow ppt shows present of carbon-oxygen double bond
- Use the melting point apparatus to measure the melting point of the prepared derivative and compare the values to the table of data

Name of carbonyl	2,4-dinitrophenylhydrazone melting point °C
acetone	126
benzaldehyde	239
butanal	123
2-butanone	118
heptanal	108
2-hexanone	110
2-methylbutanal	120
propanal	148
2-pentanone	144
3-pentanone	156

Decarboxylation of Carboxylic Acid

Removal of Carboxylate Group (COOH)
 Reagent: Soda Lime (NaOH + CaO)

- Heat
- Contain 1 less carbon
- Removal of -COOH or -COONa and replace with H



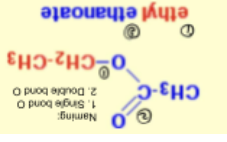
Conversion of Carboxylic Acid to Acid Derivatives

Conversion of C.A. to Acyl Chloride

- ★ ① $\text{CH}_3\text{COOH} + \text{SOCl}_2 \rightarrow \text{CH}_3\text{COCl} + \text{SO}_2^{(g)} + \text{HCl}^{(g)}$
 Better in reaction for SOCl_2 as it would only form gaseous co-product.
- ② $\text{CH}_3\text{COOH} + \text{PCl}_5 \rightarrow \text{CH}_3\text{COCl} + \text{POCl}_3^{(l)} + \text{HCl}^{(g)}$
- ③ $3 \text{CH}_3\text{COOH} + \text{PCl}_3 \rightarrow 3 \text{CH}_3\text{COCl} + \text{H}_3\text{PO}_3^{(l)}$
 Separation for acid chloride is more difficult
- HCl is an irritant → Acidic gas
- HCl observation: misty fumes
- Test for HCl: $\text{NH}_3^{(g)}$ white smoke $\text{NH}_4\text{Cl}^{(s)}$ forms

Esterification Recap

- Reflux
- Concentrated sulphuric acid (or $\text{HCl}^{(g)}$ as it provides better yield)
- Carboxylic acid and alcohol



Reaction of CA, Phenol and Alcohol with Na compound

	Carboxylic Acid	Phenol	Alcohol
Na(s)	✓	✓	✓
NaOH(aq)	✓	✓	✗
Na ₂ CO ₃ (aq)	✓	✗	✗

Halogenic Group

- Chloride atoms are highly electronegative
- Chloride have strong electron withdrawing effect and attract the electron towards itself
- Electron density of O-H bond reduces
- Acidity increase



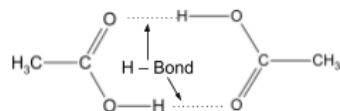
Alkyl side-chain on Acidity

- Alkyl groups are electron-donating group
- A longer alkyl chain = stronger overall effect pushing electron density towards the -COO⁻ group.
- Extra negative charge around the already negative carboxylate destabilizes it, so it is less willing to form
- Therefore the equilibrium $\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$ lies more to the left, so the acid is weaker.

Boiling Point of Carboxylic Acid

- Hydrogen bond formed between 2 carboxylic acid
- Dimer molecule
- Carboxylic acid have a higher Mr than compound with same Carbon number
- Dimer have 2 hydrogen bond (strongest intermolecular force) hence high bp

Dioic Acid



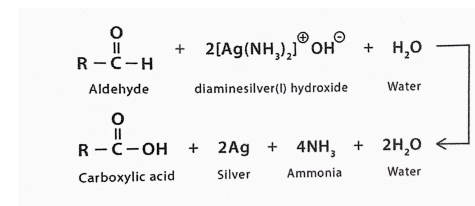
- Cause high melting point
- Good solubility in non-polar solvents

Tollen's Reagent

- To distinguish between aldehyde or ketone
- Aldehyde form silver mirror

Method:

1. Add a few drops of Tollens reagent to a clean test tube
2. Add a few drops of suspected aldehyde and warm the test tube gently
3. A silver mirror (or grey precipitate) is formed with aldehyde
4. No silver mirror with ketone

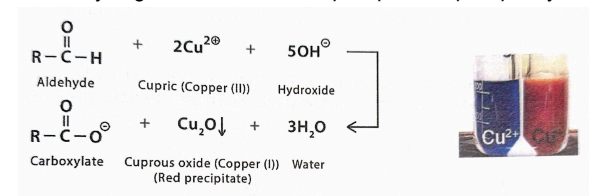


Fehling Solution (Copper (II) Sulphate and Potassium tartrate)

Similar method

A dark red precipitate is formed with an aldehydeThe blue solution remains with a ketone

All aldehyde give classic brick-red precipitate expect phenylmethane



4.5 Carboxylic Acid and their Derivatives

Test for Carboxylic Acid

- Add Sodium Carbonate
- Effervescence

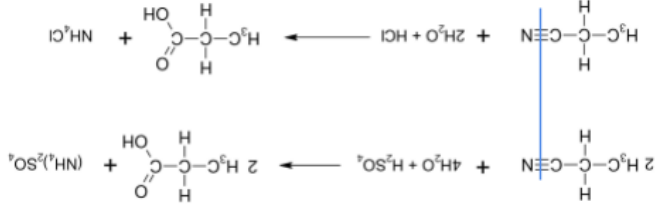
Laboratory Preparation of Carboxylic Acid

Oxidation of Primary Alcohol and Aldehyde to Carboxylic Acid

AS, A2 Alcohol, A2 Carbonyl Covered

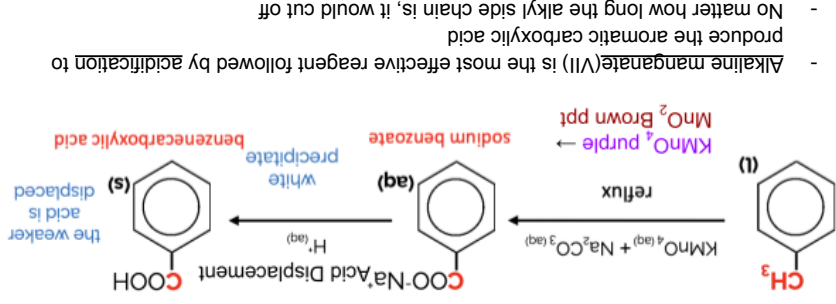
Hydrolysis of Nitrile ($\text{C}\equiv\text{N}$) to Carboxylic Acid

- $\text{C}\equiv\text{N}$ hydrolyse into COOH
- Last step of Nucleophilic Addition (Acidification) in HCN attack
- Acid Hydrolysis: Reflux with Dilute HCl or H_2SO_4 (1 step)



- Alkaline Hydrolysis: Reflux with dilute sodium hydroxide + Acid displacement of salt using H^+ to form COOH

Oxidation of Alkyl Side Chain



- Alkaline manganate(VII) is the most effective reagent followed by acidification to produce the aromatic carboxylic acid
- No matter how long the alkyl side chain is, it would cut off

Physical and Chemical Properties of Carboxylic Acid

Acidity of C.A, Phenol, Water, Alcohol

Carboxylic Acid > Phenol > Water > Alcohol

- Benzoic acid has e^- withdrawing group on $\text{C}=\text{O}$ AND $\text{O}-\text{H}$ bond is weakened
- Phenol lone pair on oxygen is delocalised into the ring AND $\text{O}-\text{H}$ bond is weakened
- Alcohol carbon chain has inductive effect (from the alkyl side chain), $\text{O}-\text{H}$ bond is strengthened so water have stronger acidic character than alcohol as $\text{O}-\text{H}$ bond is easier to be broken

Carboxylic Acid

Strength of $\text{O}-\text{H}$ bond:

- Carbonyl group ($\text{C}=\text{O}$) is an electron-withdrawing group
- Electron density in $\text{O}-\text{H}$ bond is reduced, hence weakened $\text{O}-\text{H}$ bond
- Stability of ion formed:

- Charge density on oxygen atom is spread out over the carboxylate ion
- Charge is delocalised on an electronegative carbonyl ion (conjugation)

Phenol

Strength of $\text{O}-\text{H}$ bond:

- Lone pair of oxygen is delocalised into electron ring – Mesomeric Effect
- Electron density is decreased in the $\text{O}-\text{H}$ bond
- Stability of ion formed:

- Charge density in phenoxide ion on oxygen atom is spread out over entire ion
- It is delocalised into the ring → Oxygen atom are less available for reforming $\text{O}-\text{H}$
- However, delocalisation of the charge density is on carbon atoms and not on electronegative oxygen atoms (so phenoxide ions are less stable than carboxylate ion)

Alcohol

Strength of $\text{O}-\text{H}$ bond:

- Alkyl group is electron donating group
- Increasing the electron density → This strengthens the $\text{O}-\text{H}$ bond
- Stability of ion formed:
- As there is electron donating group present (alkyl group)

- Electron density on oxygen atom is more available for bond formation with H^+
- Also alcohol does not have delocalise charge on the entire ion